

Mark(s): 1

Pointing to a photograph Preethi said "She is my father's sister's daughter Vanitha", how is Preethi's father related to Vanitha?

☐ Grandfather

☐ Cousin

☒ Uncle ✓

☐ Father

Mark(s): 1

Pointing to a photograph Preethi said "She is my father's sister's daughter Vanitha", how is Preethi's father related to Vanitha?

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☐ Father

2 ☆

Problem Solving

Mark(s): 1

For a set of 6 MCQ questions in an interview which had four choices, none of the candidate had written all correct answers and no candidates gave the same sequence of candidates who attended the interview, if everyone attempted all the 6 questions.

☒ 4095 ways ✓☐ 720 ways☐ 1296 ways☐ 360 ways

QUESTION: 3

GROUP: Aptitude

SECTION: Problem Solving

Mark(s): 1

In 10 seconds, a 125 m long train passes a man running at 5 km/hr in the same direction of the train. The speed of the train is:

☒ 50 km/hr

☐ 55 km/hr

☐ 54 km/hr

☐ 45 km/hr

QUESTION: 4

GROUP: Aptitude

SECTION: Problem Solving

Mark(s): 1

The integers a_n and b_n are defined by the relationship $(a_n + b_n\sqrt{2}) = (1 + \sqrt{2})^n$ for all integers $n \geq 1$. What is the value of $\text{HCF}(a_n, b_n)$ for all integers $n \geq 1$?

☐ $\sqrt{2}$

☐ $2^{(3n)}$

☐ Undefined

☐ 1

QUESTION: 5

GROUP: Aptitude

SECTION: Problem Solving

Mark(s): 1

I have a 60 cm long gold necklace with one joint. If I break the necklace in two pieces (by giving two independent random cuts), which of the following is closest to the

☐ 40 cm

☐ 30 cm

☐ 20 cm

☐ 25 cm

QUESTION: 6

GROUP: Aptitude

SECTION: Problem Solving

Mark(s): 1

A box contains 10 good articles and 6 with defects. One article is chosen at random. What is the probability that it is either good or has a defect?

☒ 1

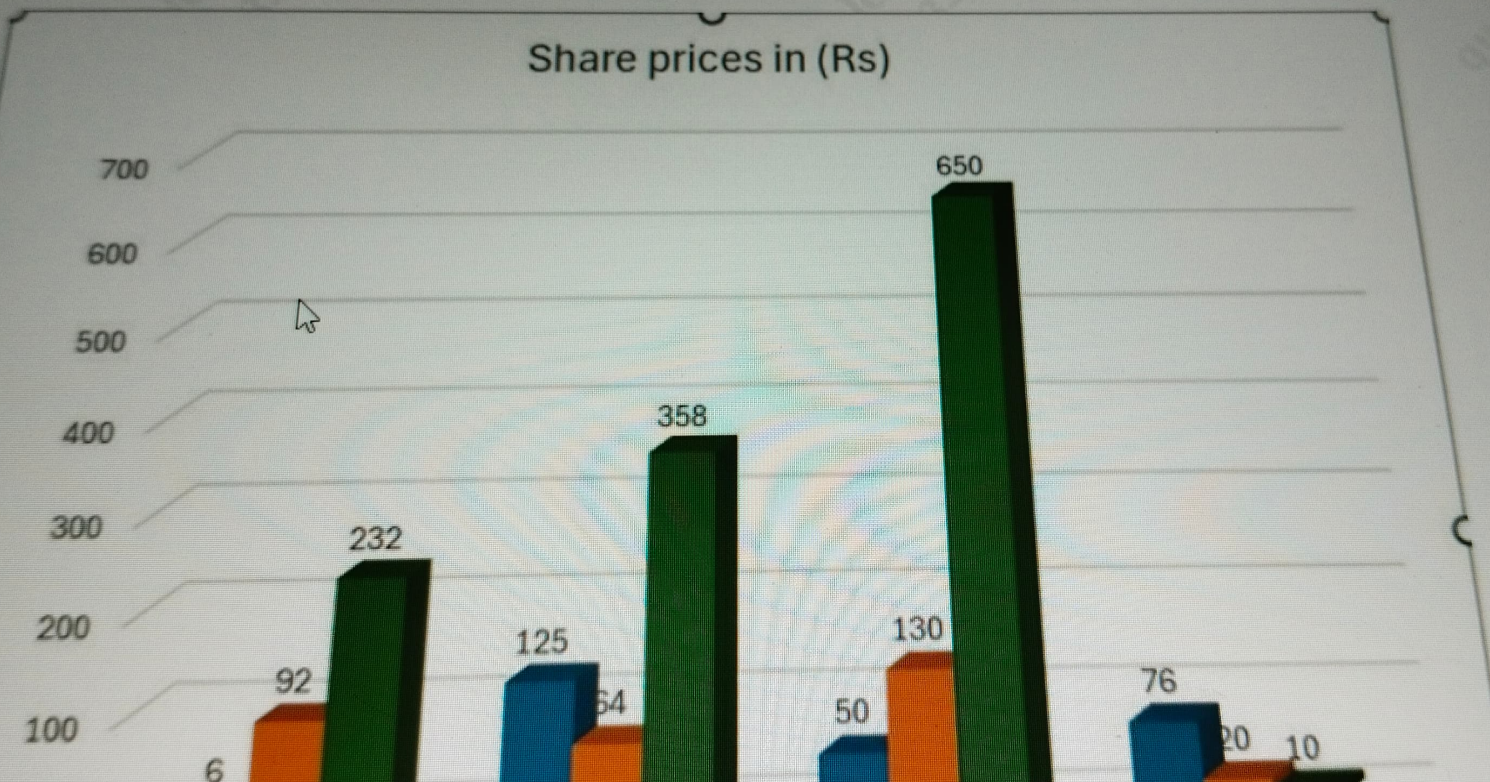
☐ $\frac{3}{8}$

☐ $\frac{5}{8}$

☐ $\frac{3}{4}$

Direction: Read the information carefully and answer the questions given below

The below bar chart gives information about the share prices (in Rs) of four shares in 3 different years.



■ 2015	92	64	130	20
■ 2020	202	358	650	10

■ 2010 ■ 2015 ■ 2020

Question: The share price of "Za" is estimated to increase by 200% from 2020 to 2022. What will be the share price of

☐ Rs. 2700

☐ Rs. 1950

☐ Rs. 1300

☐ Rs. 2400

358

650

20

10

■ 2010 ■ 2015 ■ 2020

Question: The share price of "Za" is estimated to increase by 200% from 2020 to 2022. What will be the share price of "Za" in the year 2022 as per

- ☐ Rs. 2700
- ☐ Rs. 1950
- ☐ Rs. 1300
- ☐ Rs. 2400

■ 2010 ■ 2015 ■ 2020

Question: Which share has seen decline in its price from 2010 to 2020?

☐ Za

☐ Ab

☐ Yz

☐ Xy

■ 2010 ■ 2015 ■ 2020

Question: Alex has invested Rs. 39000 in share Za in the year 2015. What is the worth of those shares in the year 2020?

☐ Rs. 36,000

☐ Rs. 7,800

☐ Rs. 78,000

☐ Rs. 1,95,000

■ 2010 ■ 2015 ■ 2020

Question: For which of the following shares, the percentage increase in the share price is maximum from 2015 to 2020?

☐ Za

☐ Ab

☐ Yz

☐ Xy

Five friends Alice, Ben, Clara, David, and Eva participate in a 10 km charity run. Each friend runs at a different constant speed. The following information is given:

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: Who among the following finishes the race first?

☒ Alice

☐ Ben

☐ David

☐ Clara

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: Who among the following has the highest running speed?

☒ Eva ✓

☐ Clara

☐ Ben

☐ David

10

- Alice completes the race in 50 minutes.

11

- Ben runs 1 kilometer in 6 minutes.

12

- Clara runs at a speed of 8 kilometers per hour.

- David takes 1 hour and 10 minutes to finish the race.

13 ☆

- Eva runs at a speed 2 kilometers per hour faster than Ben.
Question: How long does it take Clara to complete the race?

14

☐ 1 hour

15

☐ 0.83 hours

16

☐ 1.16 hours

17

☒ 1.25 hours

18

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: Who would finish the race last if Eva slowed down by 2 kilometers per hour?

☐ Ben

☐ Alice

☒ Clara

☐ David

9

Five friends Alice, Ben, Clara, David, and Eva participate in a 10 km charity run. Each friend runs at a different constant speed. The following information is given:

10

- Alice completes the race in 50 minutes.

11

- Ben runs 1 kilometer in 6 minutes.

12

- Clara runs at a speed of 8 kilometers per hour.

- David takes 1 hour and 10 minutes to finish the race.

- Eva runs at a speed 2 kilometers per hour faster than Ben.

13

Question: If David had run 1 kilometer per hour faster, how much time would he have saved?

14

☐ 0.8 hours

15 ☆

☐ 0.6 hours

16

☐ 0.12 hours

17

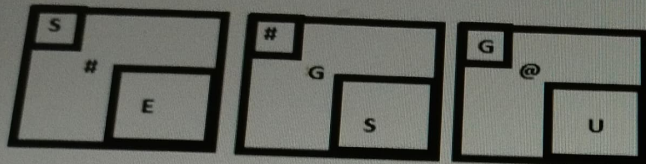
18

☐ 1.2 hours

8

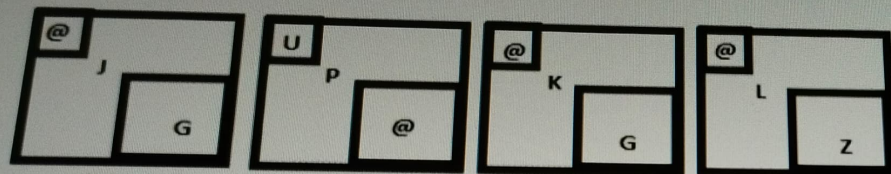
A set of problem figures and answer figures are given below, in problem figure the images are related in some way. Choose the correct figure from answer figure to complete the

9



10

11



12

13

(A)

(B)

(C)

(D)

14

☐ (B)

15

☐ (C)

16 ☆

☐ (D)

17

18

☐ (A)

7 QUESTION: 17 GROUP: Aptitude SECTION: Puzzles Mark(s): 1

8 How many pairs of letters are there in the word 'IMPURE' which has many letters between them (from both the sides) as in the alphabets from A to Z?

☐ 0

☒ 1

☐ 2

☐ 3

GROUP: Aptitude

SECTION: Problem Solving

Mark(s): 1

Daal is now being sold at Rs. 20 a kg. During last month its rate was Rs. 16 per kg. By how much percent should a family reduce its consumption so as to keep the expenditure fixed?

☒ 20%

☐ 30%

☐ 32%

☐ 22%

The marked price of a selfie stick is 100% above the cost price. When marked price is increased by 25% and selling price is increased by 25%, the profit doubles. If original marked price is Rs. 500, what is the original selling price?

☐ Rs. 550

☐ Rs. 350

☐ Rs. 450

☐ Rs. 400

QUESTION: 20 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

Mary and Maurice can do a piece of work in 10 days and 15 days respectively. They work together for 3 days and then Maurice leaves. Mary finishes the remaining work alone. In how many days is the work finished?

☐ 6 days

☐ 9 days

☒ 8 days

☐ 10 days

22

Which one of the given options is the minimum number of AND gates required for the below given?

23

 $FZ + NZ$

$$= Z(F + N)$$

24

☐ 1 ✓

25

☐ 3

26

☐ 0

27

☐ 2

22 ☆

What is the significance of the declaration "const char* p"?

23

☐ Both p and the character pointed by p are constant.

24

☒ p is a constant and not the character pointed by p. ✓

25

☐ The character referenced by p is a constant.

26

☐ The given declaration is an invalid one.

27

28

Using SJF pre-emptive scheduling algorithm, what will be the average waiting time of the processes given below?

Process	Arrival Time(ms)	Burst Time(ms)
P ₁	0.0	7
P ₂	2.0	4
P ₃	4.0	1
P ₄	5.0	4

3 ms

☐ 4 ms

☐ 5 ms

22 Which of the below given dynamic memory allocation method(s) is/are restricted, but efficient and easy to implement?

23 I) Heap allocation

24 ☆ II) Stack allocation ✓

25 ☐ Only (I)

6 ☒ Only (II) ✓

☐ Both (I) and (II)

☐ Neither (I) nor (II)

Which order of operations given in options when performed on an empty stack of size 4 will have TOP $\rightarrow$ L?

Operations

I) PUSH (A)

II) POP ()

III) PUSH (X)

IV) PUSH (Y)

V) PUSH (L)

☐ (V), (IV), (II), (I), (V), (III), (IV)

☐ (I), (III), (II), (II), (V), (II), (II)

☐ (I), (II), (III), (I), (V), (IV), (II)

☐ (II), (I), (III), (IV), (V), (II), (III)

22

Which of the scheduler given in options is also called as the job scheduler?

23

24

☐ Medium Term Scheduler

25

☐ Short Term Scheduler

26 ☆

☒ Long Term Scheduler ✓

27

☐ Intermediate Term Scheduler

28

29

Q: $f \rightarrow y \rightarrow u \rightarrow i$

What will be the content of stack after the following operations on S and Q?

PUSH(x) $\rightarrow$ to insert an element x in S

POP $\rightarrow$ to delete the top most element of S.

ADD(X) $\rightarrow$ to add an element in Q

DEL $\rightarrow$ to delete an element from Q.

I) PUSH(w), ADD(q)

II) DEL, DEL, ADD(e)

III) PUSH(DEL), PUSH(o), PUSH(u), PUSH(p)

IV) ADD(b), ADD(m), DEL, POP

IV) ADD(b), ADD(m), DEL, POP

30

31

☐ w → u → o → u

32

☐ u → o → u → p

33

☐ e → o → u → p

34

☐ w → e → o → u

35

Which of the below given statement(s) is/are correct about quick sort?

- I) Quick sort has $O(n \log n)$ time complexity in worst case.
II) Quick sort uses greedy algorithm.

☒ Neither (I) nor (II) ✓

☐ Both (I) and (II)

☐ Only (II)

☐ Only (I)

What is the resultant value after performing a NOT operation on the given binary number?

10011011

☒ 01100100 ✓

☐ 10011011

☐ 10101100

☐ 11111111

What will be the output of the program given below?

```
#include <stdio.h>

int main()
{
    static int p = 2, q = 3;

    p = p << 2 ^ q - 1;

    q = q ^ p;

    printf("%d", p & q);

    return 0;
}
```


33

☐ 5

34

☐ 6

35

☐ 8



36

☐ 10



37

38

28

What will be the output of the program given below? /* Assume address of a is 3651*/

29

```
#include <stdio.h>
```

30

```
int main()
```

31

```
{
```

```
    int a;
```

```
    int *p;
```

32 ☆

```
    int **q;
```

```
    a = 30;
```

33

```
    p = &a;
```

```
    q = &p;
```

34

```
    printf("a = %d, p = %d and q = %d", a, *p, **q);
```

35

```
    return 0;
```

```
}
```


33

```
printf("a = %d, p = %d and q = %d", a, *p, **q);
```

```
return 0;
```

34

```
}
```

35

☐ a = 30, p = 30 and q = 30 ✓

36

☐ a = 30, p = 3651 and q = 3801

7

☐ a = 30, p = null and q = null

☐ a = 3651, p = 30 and q = 3801

28

Convert the following binary number to Decimal

29

$(11111111)_2 = ?$



30

☐ $(256)_{10}$

31

☐ $(255)_{10}$



32

☐ $(355)_{10}$

33 ☆

☐ $(455)_{10}$

34

35

27

28

The leaf node of a tree is also called as

29

☐ Predecessor Node

30

☐ Internal Node

31

☐ Root Node

32

33

☒ Terminal Node



34 ☆

28

What does control flow refer to in C programming?

29

☐ The structure of a program's functions

30

☐ The flow of data in the program

31

☐ The rate of program execution

32

☒ The order in which statements are executed



28

What will be the output of the program given below?

29

```
#include <stdio.h>
```

```
int main()
```

30

```
{
```

31

```
    unsigned int f = 4, i;
```

```
    for (i = 0; i < 2; i++)
```

32

```
    {
```

```
        f <<= ++i ^ f - 1;
```

33

```
        if (f > 2)
```

```
            printf("%d ", f ^ i);
```

34

```
        else
```

```
            printf("%d ", f & i);
```

35

```
    }
```



```
printf("%d ", f & i);
```

33

```
}
```

```
return 0;
```



34

```
}
```

35

☐ 0

36 ☆

☐ 10

37

☒ 17 ✓

38

39

☐ 16

28

What is the hexadecimal equivalent of the given binary number?

29

11011100110001

30

☐ 1800h

31

☐ 9774h

32

☒ 3731h ✓

33

☐ 1589h

34

35

28

What will be the output of the program given below?

29

```
#include <stdio.h>
```

```
int main()
```

30

```
{
```

31

```
    int c[ ] = {2.8, 3.4, 4, 6.7, 5};
```

```
    int j, *p = c, *q = c;
```

32

```
    for (j = 1; j < 5; j++)
```

```
    {
```

33

```
        printf("%d ", *c);
```

```
        ++q;
```

34

```
    }
```

35

```
    for (j = 1; j < 5; j++)
```

```
{
```


32

`++p;``}`

33

`return 0;``}`

34

35

☐ 2 3 4 6 5

36

☐ 2 2 2 2 2 3 4 6

37

☐ 2.8 3.4 4 6.7 5

38 ☆

39

☐ 2 2 2 2 2 3 4 6 5

28 The given code aims to print the 5th number in an array named numbers, initialized with the values {1, 2, 3, 4, 5}. However, there is an error in the code.

```
#include <stdio.h> // Line 1
```

```
29 int main() {
```

```
30     int numbers[5] = {1, 2, 3, 4, 5}; // Line 3
```

```
31     printf("The 5th number in the array is: %d\n", numbers[5]); // Line 4
```

```
    return 0; // Line 5
```

```
32 }
```

33 ☐ Line 4



34 ☐ Line 1

35 ☐ Line 3

36

28 The given code aims to print the 5th number in an array named numbers, initialized with the values {1, 2, 3, 4, 5}. However, there is an error in the

29 `#include <stdio.h> // Line 1`

30 `int main() {`

`int numbers[5] = {1, 2, 3, 4, 5}; // Line 3`

`printf("The 5th number in the array is: %d\n", numbers[5]); // Line 4`

31 `return 0; // Line 5`

32 `}`

33 ☐ Line 4



34 ☐ Line 1

35 ☐ Line 3

28

What is the binary value of the given hexadecimal number?

29

$(A1B23)_{16}$

30

☐ $(0101\ 0110\ 0111\ 1011\ 0011)_2$

31

☐ $(1101\ 0110\ 0111\ 1011\ 0011)_2$

32

☐ $(1111\ 0110\ 0111\ 1011\ 0011)_2$

33

34

☒ $(1010\ 0001\ 1011\ 0010\ 0011)_2$ ✓

35